

AWS SES Email Sending Using PHP

1. Description

GOAL: send an email to the verified address using aws-SES with php

1. verify the recipient address inside simple email service[SES].
2. Get your smtp credentials (from SES console)
3. install php and other required dependencies in your machine
4. try to send mail by creating and executing a php file [refer aws docs for the information about the file and its contents

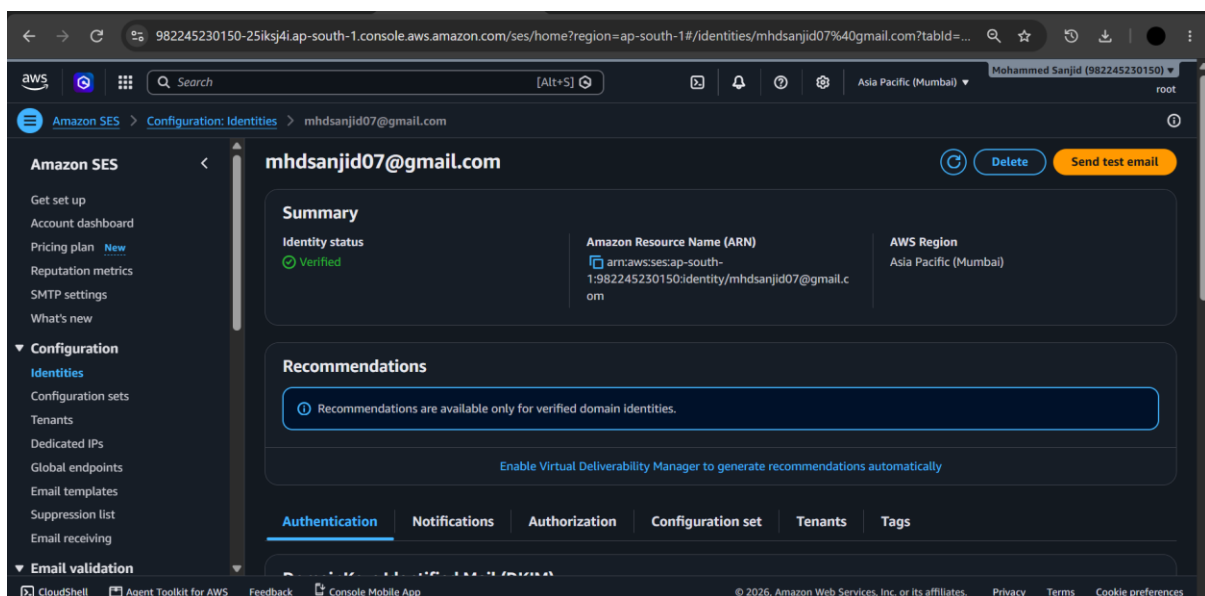
2. Environment Details

Component	Configuration
Cloud Provider	Amazon Web Services (AWS)
Region	Asia Pacific (Mumbai) – ap-south-1
EC2 Operating System	Red Hat Enterprise Linux 10.2
PHP Version	8.3.33
Composer Version	2.10.3
Email Service	Amazon SES
PHP Mail Library	PHPMailer
SMTP Port	587 (STARTTLS)

3. Procedure

Step 1: Verified the Email Identity in AWS SES

1. Opened the AWS Management Console and navigated to Amazon SES in the Mumbai region.
2. Opened the Identities section and added the email address that would be used for sending emails.
3. Completed the email verification process using the verification link received in the email inbox.
4. Confirmed that the identity status was displayed as Verified.



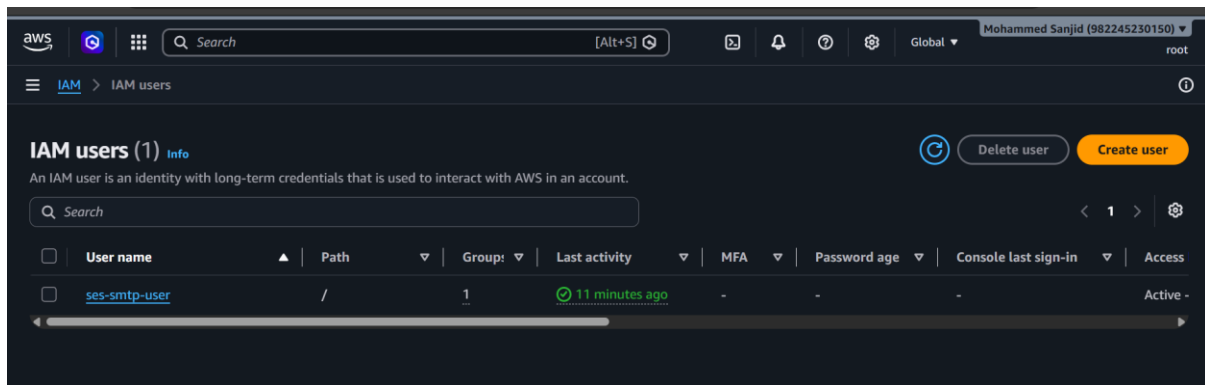
Step 2: Created SMTP Credentials

1. Navigated to the SMTP settings section in Amazon SES.
2. Selected the option to create IAM SMTP credentials.
3. Created the SMTP user and configured the required permission to send raw emails through SES.
4. Saved the generated SMTP username and password securely for use in the PHP application.

The SMTP endpoint used was:

`email-smtp.ap-south-1.amazonaws.com`

The SMTP connection was configured to use port 587 with STARTTLS encryption.



Step 3: Installed PHP on the EC2 Instance

1. Connected to the RHEL EC2 instance using SSH.
2. Checked the PHP installation using the following command:

```
php -v
```

3. Installed PHP and the required packages:

```
sudo dnf install -y php php-cli php-mbstring
```

4. Verified that PHP was installed successfully and checked the OpenSSL extension:

```
php -m | grep -i openssl
```

PHP 8.3.33 was available on the EC2 instance, and OpenSSL was enabled for secure SMTP communication.

Step 4: Installed Composer and PHPMailer

1. Installed Composer to manage PHP dependencies.
2. Verified the Composer installation:

```
composer --version
```

3. Created a project directory for the SES email application:

```
mkdir ~/ses-php  
cd ~/ses-php
```

4. Installed the PHPMailer library using Composer:

```
composer require phpmailer/phpmailer
```

5. Verified the installation and confirmed that the `vendor`, `composer.json`, and `composer.lock` files were created.

```
[ec2-user@ip-172-31-6-111 ~]$ php -v
composer --version
cd ~/ses-php
ls
PHP 8.3.33 (cli) (built: Jul 28 2026 17:56:10) (NTS gcc x86_64)
Copyright (c) The PHP Group
Zend Engine v4.3.33, Copyright (c) Zend Technologies
    with Zend OPcache v8.3.33, Copyright (c), by Zend Technologies
composer version 2.10.3 2026-08-27 13:34:23
PHP version 8.3.33 (/usr/bin/php)
Run the "diagnose" command to get more detailed diagnostics output.
composer.json composer.lock send_email.php vendor
[ec2-user@ip-172-31-6-111 ses-php]$
```

Step 5: Created and Configured the PHP Script

1. Created a PHP file named `send_email.php` inside the `~/ses-php` directory.

```
sudo vi send_email.php
```

The following PHP script was created inside the `~/ses-php` directory on the EC2 instance.

```
<?php
```

```
use PHPMailer\PHPMailer\PHPMailer;
use PHPMailer\PHPMailer\Exception;
```

```
require 'vendor/autoload.php';
```

```
$mail = new PHPMailer(true);
```

```
try {
    // SMTP configuration
    $mail->isSMTP();
    $mail->Host      = 'email-smtp.ap-south-1.amazonaws.com';
    $mail->SMTPAuth  = true;
    $mail->Username  = 'YOUR_SMTP_USERNAME';
    $mail->Password  = 'YOUR_SMTP_PASSWORD';
    $mail->SMTPSecure = PHPMailer::ENCRYPTION_STARTTLS;
    $mail->Port      = 587;

    // Sender and recipient
    $mail->setFrom('YOUR_VERIFIED_SES_EMAIL', 'AWS SES PHP Lab');
    $mail->addAddress('YOUR_VERIFIED_RECIPIENT_EMAIL');
```

```

// Email content
$mail->isHTML(true);
$mail->Subject = 'AWS SES Email Test';

$mail->Body = '<h2>Email sent successfully!</h2>
              <p>This email was sent using AWS SES and
PHP.</p>';

$mail->AltBody = 'Email sent successfully! This email was sent
using AWS SES and PHP.';

// Send email
$mail->send();

echo "Email sent successfully through AWS SES!";

} catch (Exception $e) {
    echo "Email could not be sent. Error: {$mail->ErrorInfo}";
}

```

2. Imported the PHPMailer library using Composer's autoloader.
3. Configured the SMTP connection with the SES Mumbai endpoint, SMTP authentication, STARTTLS encryption, and port 587.
4. Entered the verified sender email address, recipient email address, and the SMTP credentials generated from SES.
5. Configured the email subject and body.

The email subject was:

AWS SES Email Test

The PHP script was saved and used to send the test email through Amazon SES.

Step 6: Executed the PHP Script and Sent the Email

1. Navigated to the project directory:

```
cd ~/ses-php
```

2. Executed the PHP script:

```
php send_email.php
```

3. Initially encountered an SMTP authentication error, which was resolved by correcting the SMTP credentials.
4. Executed the script again and confirmed that the email was successfully submitted through AWS SES.

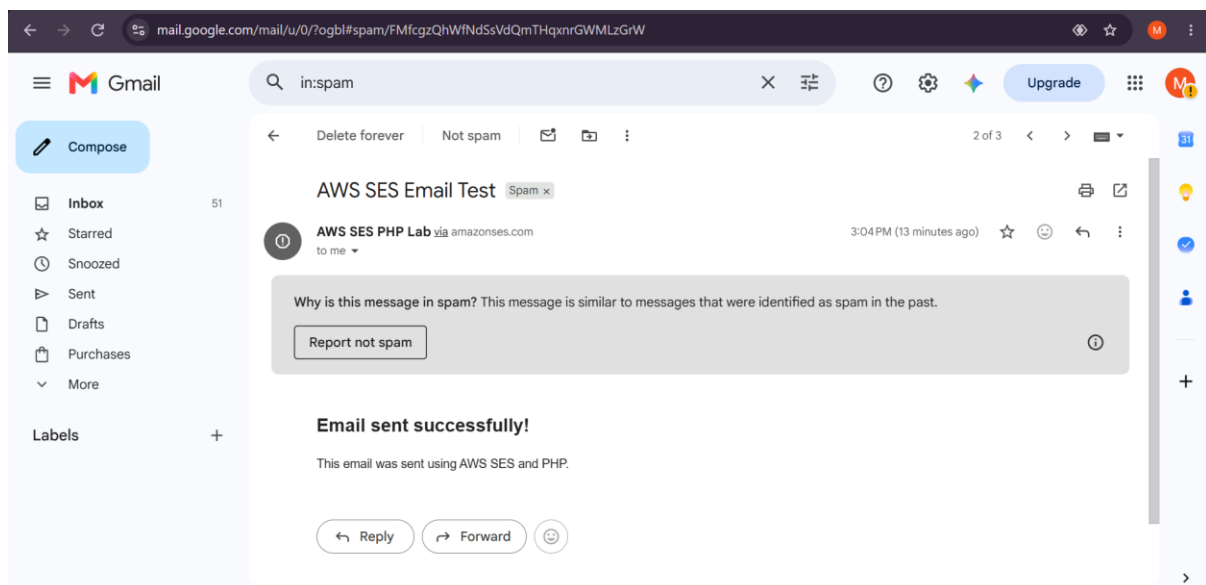
The terminal displayed the following output:

Email sent successfully through AWS SES!

```
[ec2-user@ip-172-31-6-111 ses-php]$ cd ~/ses-php
php send_email.php
Email sent successfully through AWS SES! [ec2-user@ip-172-31-6-111 ses-php]$
```

Step 7: Verified Email Delivery

1. Opened the recipient's Gmail inbox after executing the PHP script.
2. Located the email with the subject AWS SES Email Test.
3. Opened the email and verified the sender address and message body.
4. Confirmed that the email sent using PHP and AWS SES was received successfully.



4. Conclusion

Successfully configured Amazon SES to send emails using PHP from an AWS EC2 instance. Verified the email identity, created SMTP credentials, installed PHP, Composer, and PHPMailer, and configured the PHP script to communicate securely with the SES SMTP endpoint.

The PHP script executed successfully, and the test email was received in the recipient's Gmail inbox, confirming that the email-sending process was completed successfully.